



UNIVERSITY OF BUEA

**TASK
5**

CULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER
ENGINEERING

**CEF 440: INTERNET PROGRAMMING AND MOBILE
PRORAMMING**

TASK5: UI DESIGN AND IMPLEMENTATION

GROUP 19

**Design and Implementation of a Mobile App for Collection of User
Experience Data from Mobile Network Subscribers**

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Introduction

The success of a mobile application largely depends on the quality of its user interface and user experience. A well-designed interface enables users to interact efficiently with the system while ensuring accessibility, usability, and visual appeal. For the NetPulse Quality of Experience (QoE) Monitoring System, the user interface was designed to provide an intuitive platform for monitoring network performance, submitting QoE reports, and visualizing network analytics.

The application is implemented as a mobile application using React Native, TypeScript, and NativeWind (Tailwind CSS for React Native). This technology stack enables the development of a responsive, cross-platform mobile application that can run on both Android and iOS devices while maintaining a consistent user experience.

5.1 App Identity

Application Name: NetPulse

Tagline: "Measuring Network Experience, Empowering Connectivity."

Application Purpose

NetPulse is a mobile-based Quality of Experience (QoE) monitoring system designed to evaluate and visualize the performance of mobile networks from the end user's perspective. The application enables subscribers, mobile network operators, and regulators to monitor network quality through metrics such as latency, throughput, signal strength, packet loss, and user satisfaction.

Target Users

- Mobile Network Subscribers
- Mobile Network Operators (MTN, Orange, Camtel, Nexttel)
- Telecommunications Regulators
- Network Engineers and Researchers

Brand Identity

The NetPulse brand combines the concepts of network connectivity and real-time monitoring. The name reflects the continuous measurement of network performance and user experience.

Logo Concept

The NetPulse logo incorporates:

- A network signal icon representing connectivity.
- A pulse waveform symbolizing real-time monitoring.
- Blue and green color tones representing technology, reliability, and growth.



5.2 Visual Design

Design Objectives

The user interface was designed to:

- Provide a modern and intuitive experience.
- Allow users to access network information quickly.
- Present QoE metrics in a clear and understandable format.
- Maintain consistency across all application screens.

Color Scheme

Color	Purpose
Blue (#2563EB)	Primary application color
Green (#10B981)	Positive network status
Orange (#F59E0B)	Warning indicators
Red (#EF4444)	Network issues and errors
White (FFFFFF)	Background and readability
Dark Gray (#1E293B)	Text and dashboard elements

Typography

The application uses a clean sans-serif font to improve readability on mobile devices.

Font characteristics:

- Simple and modern appearance
- High readability on small screens
- Consistent spacing and hierarchy

User Interface Components

The application consists of the following major screens:

Login Screen

- User authentication
- Secure account access
- Role-based authorization

Subscriber Dashboard

- QoE score display
- Network test submission
- Personal performance history
- Real-time network statistics

Operator Dashboard/ Administrator Dashboard

- Network performance monitoring
- Regional QoE analytics
- Subscriber reports
- Performance comparison charts
- National network overview
- Operator comparisons
- Regulatory monitoring
- System management tools

Regulatory Dashboard

- National network overview
- Operator comparison statistics
- QoE compliance monitoring
- Historical performance reports

User Experience Considerations

The interface was designed using the following principles:

- Simplicity
- Accessibility
- Responsiveness
- Consistency
- Minimal user effort
- Fast navigation

Navigation Design

The application uses bottom-tab navigation to improve usability on mobile devices.

Main navigation tabs include:

- Home
- Reports
- Analytics
- Notifications

This structure minimizes navigation complexity and improves user experience.

5.3 Frontend Implementation

Development Framework

The frontend of NetPulse was developed using:

React Native

React Native was selected because it enables cross-platform mobile development using a single codebase.

TypeScript

TypeScript provides static typing, improved code quality, and easier maintenance of large-scale applications.

NativeWind

NativeWind enables the use of Tailwind CSS utility classes within React Native, resulting in faster and more consistent UI development.

Expo

Expo simplifies mobile application development, testing, and deployment.

React Navigation

React Navigation manages screen transitions and navigation between application modules.

Application Architecture

The frontend follows a component-based architecture.

Major components include:

- Authentication Components
- Dashboard Components
- Analytics Components
- Network Testing Components

This architecture improves maintainability, scalability, and code reusability.

State Management

React Hooks are used to manage application state.

Key hooks include:

useState()

Used for:

- User authentication state
- QoE report management
- Theme preferences
- Dashboard data

useEffect()

Used for:

- API data fetching
- Session management

- Real-time dashboard updates

User Authentication

The authentication module supports:

- User registration
- Secure login
- Session persistence
- Role-based access control

User roles include:

- Subscriber
- Operator
- Administrator

Each role is automatically redirected to the appropriate dashboard after successful authentication.

QoE Data Visualization

The application presents QoE information using:

- Summary Cards
- Performance Indicators
- Interactive Charts
- Regional Comparison Tables
- Operator Ranking Views

These visualizations help users interpret network performance data quickly and effectively.

Responsive Mobile Design

The application is optimized for:

- Android smartphones
- Android tablets

- iOS smartphones
- iOS tablets

Responsive layouts are implemented using Flexbox and NativeWind utility classes.

Error Handling and User Feedback

To improve reliability and usability, the application provides:

- Loading indicators
- Error notifications
- Retry mechanisms
- Success confirmations

These features ensure a smooth user experience even when network issues occur.

Frontend Security Considerations

The frontend implementation includes:

- Secure authentication workflows
- Protected dashboard access
- Session management
- API request validation
- Role-based authorization

These measures help protect user information and system resources.

Theme Support

NetPulse supports:

- Light Mode
- Dark Mode

This improves usability and accessibility under different viewing conditions.

Language Support

Our application supports :

- English
- French

Demo Video

A demonstration video of the NetPulse mobile application has been prepared to showcase the functionality and user experience of the system.

The demo video illustrates the following features:

- User authentication and login process.
- Navigation through the subscriber dashboard.
- Submission of Quality of Experience (QoE) reports.
- Visualization of network performance metrics.
- Operator dashboard functionalities.
- Administrative monitoring and analytics features.
- Real-time data visualization and reporting.

The video provides a practical demonstration of the application's workflow and highlights how the system can be used to monitor and analyze mobile network Quality of Experience.

Conclusion

The NetPulse mobile application's user interface was designed to provide an intuitive, responsive, and visually appealing experience for monitoring Quality of Experience (QoE) metrics. By leveraging React Native, TypeScript, NativeWind, Expo, and React Navigation, the system delivers a modern cross-platform solution capable of serving subscribers, mobile network operators, and telecommunications regulators. The implementation emphasizes usability, scalability, accessibility, and efficient visualization of network performance data, making NetPulse a practical and effective QoE monitoring platform.